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| **Fiscal Year:** | FY 2012 | **Task Last Updated:** | FY 05/16/2012 |
| **PI Name:** | Park, Sung-Yong Ph.D. | | |
| **Project Title:** | Heat Transfer on Multiscale Textures with Electrowetting Control | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FLUID PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Fluid physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | spark@teledyne.com | **Fax:** | FY |
| **PI Organization Type:** | INDUSTRY | **Phone:** | 805-373-4269 |
| **Organization Name:** | Teledyne Scientific Company | | |
| **PI Address 1:** | Thermal and Flow Physics | | |
| **PI Address 2:** | 1049 Camino Dos Rios | | |
| **PI Web Page:** | | | |
| **City:** | Thousand Oaks | **State:** | CA |
| **Zip Code:** | 91360-2362 | **Congressional District:** | 24 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2008 Fluid Physics NNH08ZTT002N |
| **Start Date:** | 12/01/2011 | **End Date:** | 10/01/2012 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | 3 |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Chao, David | **Contact Phone:** | 216-433-8320 |
| **Contact Email:** | david.f.chao@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 10/1/2012; okayed per D. Chao/GRC (Ed., 8/30/12) | | |
| **Key Personnel Changes/Previous PI:** | Sung-Yong Park is the new PI effective in FY2012. Previous PI Dr. Chung-Lung Chen retired from Teledyne in November 2011. | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNC09CA36C | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | NOTE: PI change for FY2012. See Chung-Lung Chen (PI) and project titled "Heat Transfer on Multiscale Textures with Electrowetting Control" for previous reports--FY2009-FY2011.<br>Modern electronic systems used in space must be reliable and efficient with thermal management unaffected by outer space environments. Heat pipes and vapor chambers based on phase-change heat transfer mechanisms have proven to be an efficient approach to dissipate heat in many space or ground-based thermal-energy systems. However, a layer of capillary wick structure fully covering the chamber walls, e.g., condenser and evaporator surfaces, creates additional thermal resistance, because the effective thermal conductivity is much lower in saturated wicks. This thermal barrier can generate an unacceptable temperature rise, especially in high heat-flux applications. | | |

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| **Task Description:** | To overcome this issue, we propose to develop an innovative wickless and electrowetting-controlled vapor chamber with bioinspired two-tier textures on the chamber wall. These engineered superhydrophobic / superhydrophilic multiscale textures, formed by growing carbon nanotubes (CNTs) on top of micropillars on a silicon chip, mimic the structure of lotus leaves; they can promote dropwise condensation at the condenser site, and enhance film evaporation at the evaporator site. The condensate droplets will be dislodged from the condenser site and circulated to the heat source via electrowetting transport. Droplet transport by electrowetting without any mechanical moving parts has great potential in thermal management due to its programmable and reconfigurable characteristics, prompt response, and low power consumption. Most importantly, given a two-dimensional array of control electrodes, condensate can be transported over the low-friction hydrophobic surface to a hot spot without the significant pressure loss that is associated with fine pore wicks. The transported droplets will be captured by a few finger-like extensions from the heat source, which are hydrophilized by a coating of UV&Ozone-treated carbon nanotubes. This hydrophilic coating will force droplets to cling to the hot spot surface, changing heat transfer into the more efficient form of thin film evaporation.<br><br>In this program, we aim to a) investigate wetting characteristics and phase-change heat transfer on multiscale textures; b) quantify electrowetting transport of condensates and transient heat transfer on the textured surface; c) develop numerical models of heat transfer on the textures. Reduced gravity flight tests will be performed to validate our analysis and ground tests.<br><br>This innovative design from this program will enable a new generation of high-performance space integrated systems to operate at high-power densities without problems from temperature gradients, increased weight, or added complexity. Low-pressure-loss droplet transport on multiscale textures, combined with adaptable, low-power electrowetting, has the potential to significantly improve vapor chamber efficiency, increase its robustness and lifetime, and reduces the mass of space flight systems needed to support future human space exploration activities. The application of nanothermal materials and adaptive liquid transport technology based on electrowetting will lead to a major breakthrough in the thermal management field. The success of the proposed effort can considerably enhance thermal management in the future space explorations. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | One of the major applications of phase-change heat transfer at low-gravity conditions is the design and maintenance of robust and efficient thermal control systems for future space exploration and satellite applications. Heat pipes and vapor chambers based on phase-change heat transfer mechanisms have proven to be good candidates to dissipate heat in many space thermal-energy systems. However, a layer of capillary wick structure fully covering the heat pipe and vapor chamber inner walls creates additional thermal resistance because the effective thermal conductivity is much lower in saturated wicks. This thermal barrier can generate an unacceptable temperature rise, especially in high heat-flux and harsh-environment applications.<br>To address this issue, Teledyne Scientific & Imaging Company (TS&I) proposes to develop an innovative wickless and electrowetting-controlled vapor chamber with bioinspired two-tier textures on the chamber wall. This innovative design will enable a new generation of high-performance space integrated systems to operate at high-power densities without problems from temperature gradients, increased weight, or added complexity. Low-pressure-loss droplet transport on multiscale textures, combined with adaptable, low-power electrowetting, has the potential to significantly improve vapor chamber efficiency, increase its robustness and lifetime, and reduce the mass of space flight systems needed to support future human space exploration activities. The application of nanothermal materials and adaptive liquid transport technology based on electrowetting will lead to a major breakthrough in the thermal management field. The success of the proposed effort can considerably enhance thermal management in the future space explorations. |
| | Microelectronics technologies have dramatically improved with reduced feature size, fast switching speed, and increased package density. These advances result in increased heat generation that may cause performance degradation or eventual device failure due to thermal stress. Therefore, effective thermal rejection techniques are critically demanded to prevent from overheating. Recently electrowetting-on-dielectric (EWOD) has emerged as one of the effective on-chip cooling techniques, due to its unique characteristics such as extremely low power consumption for fluid transport, no need of high pressure head encountered in continuous flows confined in microchannels, fast response time, and easy implementation. Most importantly, EWOD allows rapid droplet transportation along any arbitrary path to the target heat sources where heat flux concentrated can be effectively controlled.<br>Conventional EWOD devices are typically implemented by sandwiching droplets between two parallel plates and droplet actuation is achieved by addressing digitized electrodes. Various droplet manipulation functions such as droplet injection, splitting, transportation, and mixing have been demonstrated on a sandwiched configuration. An issue of this sandwiched EWOD structure for a cooling application is the fact that vaporization of coolant droplets is not allowed in such a closed configuration, resulting in very limited heat rejection only relying on sensible heat transfer. Recently, we demonstrated a coplanar EWOD technique that enabled two-phase heat transfer on an open-plate surface and thus cooling efficiency is significantly enhanced. Although careful droplet control is very critical to achieving effective cooling, important droplet manipulation functions such as injection and splitting with precise volume control has not been achieved in coplanar EWOD. To successfully complete a droplet cutting process, one critical requirement is the constraint of droplet height, which can be provided in a sandwiched EWOD. However, coplanar EWOD can’t provide such a constraint for flexible droplet manipulations. To overcome the issue of flexible droplet manipulation, we have developed a single-sided EWOD (SEWOD) device that allows two-phase heat transfer on a single-plate configuration for effective hot spot cooling, and completes important droplet manipulation functions such as continuous droplet injection, rapid transport, and droplet position on hydrophilic hot spots.<br><br>We have developed the new electrical control system utilizing National Instrument’s LabVIEW software to control the timing and regulation of the excitation voltage which was used for determining the droplet volume and dispensing frequency. For rapid transport of a droplet, two adjacent electrodes were activated simultaneously - one is powered and another is grounded. This paired electrode activation sequentially applied to the next pair of the electrodes to achieve continuous droplet transportation. With the LabVIEW-based pair control scheme, we have achieved important droplet manipulation functions necessary for effective two-phase cooling on SEWOD. Those droplet handling functions include continuous droplet injection, rapid transport, and droplet positioning onto the target heat source. On a SEWOD surface, droplets can be continuously injected with precise volume control less than 1% variation and rapidly transported to the target hot spots at a speed as fast as 7.5 cm/s. We have also studied hot spot surface effect on cooling, because most heat |

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| **Task Progress:** | transfer studies using conventional sandwiched EWOD and coplanar EWOD technologies have been conducted only on the hydrophobic hot spot surface. For this study, we have prepared two SEWOD devices with different hot spot surface properties: one was a hydrophobic surface provided by spin coating of a FluoroPel polymer solution and another was a hydrophilic surface on a thin Ta2O5 film deposited by an e-beam evaporator. The surface properties were examined by measuring its contact angle. The droplet sits on the hydrophobic surface with the contact angle of 118°, while the angle was measured to be 68° on the hydrophilic Ta2O5 coated surface.<br><br>Our cooling tests indicate that the hydrophilic surface property was able to effectively remove heat from the hot spot. On the hydrophobic hot spot surface, it was rarely positioned on the hot spot area. Due to low friction or less pinning effect on the hydrophobic surface, the droplet skittered away and thus the hot spot area was partially covered by the coolant droplet. During evaporation process, explosive vapor steam moves the droplet away from the target hot spots, resulting in a quick temperature increase. The hot spot initially heated up to 163.5°C and it experienced a quick but small temperature drop (delta T) about 21°C, when a 5 µL water droplet was transported to the hydrophobic hot spot location. During a droplet evaporation process, the temperature at the hot spot was maintained around 140°C. After a droplet was fully vaporized, the hot spot temperature was quickly recovered. This cooling study was repeated on the hydrophilic hot spot surface to see how the hot spot surface property has an influence on heat transfer. Similarly the hot spot was initially heated up to 165°C and a 5 µL droplet was delivered to the target location. On the hydrophilic hot spot surface, the droplet was firmly stuck and expanded the contact area. The temperature greatly dropped down to 115°C, (delta T>50°C). The hot spot was maintained below 110°C. After the droplet evaporation is fully developed, the surface temperature at the hot spot increases. One interesting observation is the shorter duration of droplet vaporization on the hydrophilic surface. This indicates that the heat dissipation rate is much faster on the hydrophilic surface than the hydrophobic. Our studies conclude that the hydrophilic surface modification on the target heat sources can be very useful to improve heat transfer capability on SEWOD. Not only can a hydrophilic surface make a droplet firmly stick onto the hot spot surface even during explosive vaporization, but also increase the heat transfer area by expanding the droplet contact area<br><br>To stably remove heat generated from a heat source without overheating and droplet flooding, it is necessary to continuously feed droplets with precisely controlled volume as well as frequency. Four water droplets are successively delivered to the hydrophilic hot spot, which was heated up to 172°C under a heat flux density of 6.9 W/cm2. When the first 30 µL water droplet arrived at the hydrophilic hot spot, the temperature instantly dropped down to 89.5°C (delta T=81.5°C) within 2s and was maintained less than 100°C for 106s. At the moment of completing the droplet vaporization, the next droplet was quickly delivered to avoid overheating. This process was continued and the hot spot temperature was able to be mitigated below 100°C for 475s using four successive droplets.<br><br>We have been further developing LabVIEW programming for a feedback control of the hot spot temperature. This control scheme enables a real-time temperature monitoring that will be used to control injected droplet volume and feeding frequency to mitigate the hot spot temperature below a certain temperature. Our SEWOD-driven cooling technique promises to potentially function as a wickless vapor chamber or heat pipe with enhanced thermal managing capabilities. |
| **Bibliography Type:** | Description: (Last Updated: 09/09/2022) |
| **Abstracts for Journals and Proceedings** | Park S-Y, Cheng J, Chen C-L. "Active hot spot cooling controlled by single-sided electrowetting-on-dielectric (SEWOD)." ASME 2012 Summer Heat Transfer Conference, Puerto Rico, July 8-12, 2012.<br>Proceedings of the ASME 2012 Summer Heat Transfer Conference, Puerto Rico, July 8-12, 2012. ASME Abstract HT2012-58111. (In press as of May 2012) , May-2012 |